```
[root@secOps analyst]# echo NguyenPhiTruong
NguyenPhiTruong
[root@secOps analyst]# netstat -tunpa
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
  PID/Program name
tcp        0      0 0.0.0.0:6666            0.0.0.0:*               LISTEN
  2166/nginx: master
```

***netstat -tunpa*:** This command lists all active network connections and listening ports on the system. In this step, it is used to verify that the malicious web server (nginx) has started successfully and is listening for connections on TCP port 6666.

```
[root@secOps analyst]# echo NguyenPhiTruong
NguyenPhiTruong
[root@secOps analyst]# tail -f /var/log/snort/alert
03/04-07:33:21.303264  [**] [1:1000003:0] Malicious Server Hit! [**] [Priority:
0] {TCP} 209.165.200.235:41904 -> 209.165.202.133:6666
```

tail -f /var/log/snort/alert: The tail command with the -f option continuously monitors and outputs the newest lines added to a file.

```
[root@secOps analyst]# echo NguyenPhiTruong
NguyenPhiTruong
[root@secOps analyst]# iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

    0     0 DROP       tcp  --  any    any     anywhere             209.165.202.
133      tcp dpt:6666

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
```

iptables -L -v: This command lists (-L) all the current firewall chains and rules with verbose (-v) details, including packet and byte counts. It is used to provide visual proof that the rule designed to DROP traffic to the malicious server has been successfully added to the FORWARD chain.

1. What are the main differences between how Firewalls and intrusion detection systems (IDS) inspect network traffic? Why is it important to deploy both?

- Firewalls usually compare the packet header against a rule set, while IDSs often use the packet payload for rule set comparison. It is important to deploy both for layered defense: firewalls block unauthorized connections, while IDSs detect malicious content hidden within the allowed connections.

2. Discuss how an attacker might attempt to bypass both IDS and Firewall detection. What techniques could be used, and how might a network defender respond?

- Attackers might use encryption to hide payloads from the IDS, packet fragmentation to evade signature matching, or IP spoofing to bypass firewall rules. Defenders can respond by enabling SSL/TLS decryption, configuring devices to reassemble fragmented packets, and applying strict anti-spoofing rules.

3. Explain the concept of "default deny" in firewall configurations. What are its advantages and potential drawbacks in a production environment?

- **Default deny is a design that drops all packets by default while manually specifying what traffic should be allowed. The advantage is protecting the network from unknown protocols and attacks. The drawback is that missing an "allow" rule can disrupt legitimate services.**